

Ex. No: 8	Working With Node.js + MySQL
06.04.2026	

Theme: Online event management system

Aim:

To design and develop a web-based event management system using **Node.js** and **Express**, which enables users to register for events through an HTML interface, stores and retrieves registration data using a **MySQL database**, and allows an administrator to view and search registration details efficiently.

Description:

This project focuses on developing a web-based event management system using Node.js and Express. It provides a user-friendly interface where users can register for various events through an HTML form. The submitted data is processed on the server side and stored securely in a MySQL database. The system ensures proper validation of user inputs before saving the data. It also allows retrieval of stored information, enabling efficient data management. An administrator can view recent registrations and access participant details when needed. Overall, the application demonstrates full-stack development by integrating frontend, backend, and database technologies effectively.

Code:

Server.js

```
// =====  
// IMPORTS  
// =====  
  
const express = require("express");  
require("dotenv").config();  
const mysql = require("mysql2");  
const cors = require("cors");  
  
const app = express();
```

```

// =====
// MIDDLEWARE
// =====

app.use(cors());
app.use(express.json());

// =====
// DATABASE CONNECTION
// =====

const db = mysql.createConnection({
  host: process.env.DB_HOST,
  user: process.env.DB_USER,
  password: process.env.DB_PASSWORD,
  database: process.env.DB_NAME
});

db.connect(err => {
  if (err) {
    console.log("DB Error:", err);
  } else {
    console.log("MySQL Connected ");
  }
});

// =====
// ROUTES
// =====

// Test route
app.get("/", (req, res) => {
  res.send("Server running ");
});

// Save registration
app.post("/register", (req, res) => {
  const { name, email, event } = req.body;

  const sql = "INSERT INTO registrations (name, email, event) VALUES (?, ?, ?)";

  db.query(sql, [name, email, event], (err) => {
    if (err) {

```

```

        console.log(err);
        return res.status(500).send("Error saving data");
    }
    res.send("Registration saved successfully ✅");
  });
});

// Get latest 6 registrations
app.get("/recent", (req, res) => {
  const sql = `
    SELECT name, event, created_at
    FROM registrations
    ORDER BY created_at DESC
    LIMIT 6
  `;

  db.query(sql, (err, results) => {
    if (err) {
      console.log(err);
      return res.status(500).send("Error fetching data");
    }
    res.json(results);
  });
});

// =====
// START SERVER
// =====

app.listen(3000, () => {
  console.log("Server running on port 3000 🚀");
});

```

Input & Output:

```

E:\aiwd\backend>node server.js
◇ injected env (4) from .env //
Server running on port 3000 🚀
MySQL Connected ✅
|

```

View Registrations

santeep

Corporate Networking

6/4/2026, 7:08:00 pm

hariram

Technical Summit

6/4/2026, 7:01:50 pm

hjcj

Technical Summit

6/4/2026, 6:59:30 pm

Raj

suriya

Elgin

The screenshot shows a MySQL database management interface with a tab labeled 'hospitalmanagment_db*'. The SQL editor contains the following queries:

```
7      event VARCHAR(100),
8      created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
9  );
10 •  SHOW TABLES;
11
12 •  SELECT * FROM registrations WHERE name = 'santeep';
13
14
```

The 'Result Grid' is displayed below the queries, showing the results of the last query. The grid has columns: id, name, email, event, and created_at. The first row shows a registration for 'santeep' with ID 6, email 'santeep@karunya.edu.in', event 'Corporate Networking', and timestamp '2026-04-06 19:08:00'. A second row with all NULL values is also visible.

id	name	email	event	created_at
6	santeep	santeep@karunya.edu.in	Corporate Networking	2026-04-06 19:08:00
NULL	NULL	NULL	NULL	NULL

Result:

The developed web-based event management system successfully enables users to register for events through an interactive interface. The application efficiently stores and retrieves registration data using a MySQL database. It provides accurate display of recent registrations, ensuring real-time data handling. Overall, the system meets the intended objectives of integrating frontend, backend, and database operations effectively.